

AHNAF ZAREEF

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EDUCATION

McMaster University

Bachelor of Computer Engineering, (GPA: 3.5/4.0)

Hamilton, ON

Sep. 2024 – April 2028

EXPERIENCE

Firmware/ML Developer Intern

May 2026 – Present

Carobot Learning & Research Organization

Toronto, ON

- Built firmware and host tooling for on-device image classification on an **ESP32-S3**, running inference fully offline.
- Implemented **C firmware** for live camera capture while receiving weights, enabling training and flashing over USB.
- Used **FreeRTOS** to split MicroPython and UVC camera capture into separate tasks, preventing streaming from blocking training and inference.
- Evaluated microcontroller variants, ruling out models for limited PSRAM and USB support, and benchmarked MobileNet V1 at **2.1x** available device RAM, pivoting to smaller models trained on raw images to fit on-device.

Power & Data Acquisition Member

Sep. 2025 – Present

McMaster Solar Car Project

Hamilton, ON

- Designed IV Curve Tracer PCB in **Altium** using a Darlington pair for high-gain current amplification, ESP32 and a Hall effect sensor to characterize per-cell efficiency and detect faults.
- Developed protection circuitry: MOSFET switching on cell output, and Zener clamping on the ADC preventing overvoltage.
- Wrote C firmware implementing an IV sweep with averaging, settling delays and offset calibration for more accurate readings, reducing noise by **30%**.

PROJECTS

LCD FPGA Driver

- Wrote a **C driver** for an **ILI9341** SPI display on a **MicroBlaze** soft-core CPU, communicating over AXI SPI and GPIO and fitting the entire system in on-chip BRAM to run without external memory.
- Built a graphics library for the display, supporting shape fills, text, and images without framebuffer to fit on-chip memory.
- Added resistive touch input with calibration, mapping sensor readings to screen coordinates for precise input.
- Packaging the LCD driver as a reusable AXI IP core for integration into other Vivado designs.

I2C Master Controller

- Designed a **Verilog** I2C master controller to interface a processor with sensors and peripherals over the I2C bus.
- Implemented open-drain signaling and start/stop framing so devices safely share the bus without conflicts.
- Verified the design with a testbench and slave model, then deployed it as a memory-mapped peripheral on a MicroBlaze SoC with a C driver on the **Arty S7-25**.

Hardware Accelerator

- Built a binarized neural network accelerator on an FPGA, replacing costly multiplication with simple bitwise logic to achieve **32x** model compression.
- Hit **95%** accuracy within FPGA's logic budget by processing the network sequentially by layer rather than in parallel.
- Reshaped weight storage to match the FPGA's BRAM geometry, cutting memory **47%** to fit 270K weights.

Lidar Mapper

- Built C firmware for a spatial scanner, rotating a time-of-flight distance sensor with a stepper motor over I2C to sample 128 points per full 360° rotation.
- Added start/stop control and range validation, streaming the distance and angle data over UART to **MATLAB** to reconstruct a 3D point cloud.

TECHNICAL SKILLS

Languages: C, C++, Verilog, Python, Bash

Protocols: I2C, SPI, UART, AXI, CAN, USB, Ethernet

Tools: Vivado, Vitis, Quartus, MicroBlaze, Altium, ESP-IDF, LTSpice, PSpice, MATLAB, TinyUSB

Platforms: Xilinx, ARM Cortex, ESP32, DE10-Lite, FreeRTOS